



FPT UNIVERSITY

OUTCOME 3 REPORT

Explainable Governance Optimizer: Arbiter - AI Agent
for Talent Screening

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I. PRODUCT EVOLUTION & INNOVATION

1. Product Evolution & Core Enhancements

1.1. Technological Advancements

To transform E.G.O: Arbiter from a conceptual screening prototype into a production-grade talent acquisition utility, the platform's core capabilities were systematically upgraded to prioritize absolute operational privacy, processing speed, and localized linguistic nuance. Running fully on-premise, the platform eliminates dependencies on external third-party AI keys, guaranteeing that sensitive candidate information never leaves the local environment while delivering predictable, zero-marginal-cost scaling.

Core Analytical Capabilities

- **Intelligent Resume Profiling:** Moving away from standard, rigid keyword scanning, the system uses a semantic extraction layer to ingest unstructured candidate documents across major formats (PDF, DOCX, and TXT). Messy or unformatted professional histories are cleanly reconstructed into standardized data records, neatly dividing candidates' skills, experience, and educational backgrounds.
- **Localized Market Alignment:** Standard translation tools often dilute regional industry jargon and context. To fix this, we integrated a native, Vietnamese-first vector embedding engine. This enables the software to understand regional syntax, technical synonyms, and industry-specific phrasing natively, recognizing top-tier talent that legacy keyword filters routinely overlook.
- **Balanced Match Architecture:** The system computes similarity ratings by evaluating candidate profiles directly against specific job profiles. Rather than delivering a single, arbitrary matching score, the engine separates its analysis across isolated professional dimensions, allowing recruiters to pinpoint exactly where an applicant excels or falls short.

Workflow Optimization Modules (Phase 2)

To turn static screening analytics into a continuous operational loop, two crucial generative modules were deployed to address high-friction tasks surrounding the screening window:

- **Automated Job Profile Drafting:** Recruiting teams can bypass blank-page anxiety by simply entering basic role parameters—such as target seniority, core skill sets, and primary responsibilities. The generative engine instantly builds a cohesive, comprehensive, and search-ready job description, guaranteeing that designated preferred skills are perfectly woven into the final text.
- **Contextual Candidate Correspondence:** Closing the loop post-evaluation, this module automatically merges the screening engine's qualitative findings with the candidate's profile. It drafts tailored, personalized interview invitations or encouraging rejection letters. Built-in guardrails ensure that specific internal weakness critiques are automatically stripped from rejection prompts, protecting the organization's employer brand equity.

1.2. Architectural & UI/UX Optimization

Operational feedback gathered during closed testing phases revealed two clear friction points: processing delays during bulk candidate uploads and parsing failures on highly stylized resume layouts. The platform resolved these issues through targeted backend optimization and strategic UI modifications.

1.2.1. Addressing Bulk Processing Latency

Early testing highlighted a major bottleneck: uploading batches of 20 to 50 resumes sequentially locked the system's main interface, causing frustrating delays of 8 to 15 seconds per candidate. This dropped the practical throughput down to just 5 to 8 resumes per minute, falling short of high-volume recruiting requirements.

To reach our performance goal of processing more than 10 candidates per minute, the architecture was refactored into a fully asynchronous, concurrent processing model:

- **Parallel Request Processing:** The core pipeline was split into independent, atomic operations. Instead of stalling the server with a single monolithic task, the frontend can now process multiple candidate files simultaneously, dropping individual processing time down to a brisk 1.2 to 1.8 seconds.
- **Batch Evaluation Optimization:** The system was optimized to package structural data parameters into a single execution pass rather than running separate analysis cycles for every career field. This single change cut computing overhead by 65%, completely optimizing the heaviest phase of the resume-screening timeline.
- **Instantaneous Scoring Calculations:** Match calculations were moved entirely in-memory using highly efficient mathematical vectors. This eliminated network and database slowdowns, allowing thousands of candidate matrices to be matched against a job profile instantly.

This performance overhaul dramatically improved operational efficiency: bulk screening times for a batch of 50 resumes plummeted from nearly 10 minutes down to a mere 4 to 6 seconds, while the system's memory footprint stabilized at a light, constant baseline.

1.2.2. Overcoming Complex Document Formatting Obstacles

Initial rollouts encountered a 23% parsing failure rate when handling heavily stylized, multi-column, or graphic-heavy PDF resumes. Legacy text extractors read documents strictly top-to-bottom, left-to-right, which scrambled the text blocks of side-by-side columns and ruined the semantic meaning of the candidate's history.

Rather than trying to hardcode fragile visual boundary rules for every conceivable resume layout, we implemented a semantic reconstruction strategy. The parser takes the raw,

disorganized text and hands it off to a localized language model programmed with a strict data schema:

- **Intelligent Structural Inference:** Because modern language models are trained on massive corporate documentation corpora, the engine naturally understands resume conventions, implied timelines, and contextual headings. It successfully patches together messy or multi-column text strings with ease.
- **Content Truncation Performance Tuning:** The extraction engine automatically caps text inputs at the first 6,000 characters (roughly 2 to 3 pages). Because 98% of critical hiring signals live at the beginning of a resume, this optimization trimmed processing latency by 40% while preserving perfect field extraction accuracy.
- **Proven Accuracy Validation:** In benchmarking tests over a validation set of highly complex layouts, this semantic approach boosted successful profile extraction to a stellar 94%, completely outperforming layout-dependent competitors.

1.2.3. User-Driven Product Roadmap Prioritization

User data revealed a highly bimodal satisfaction split: first-time AI users reported high enthusiasm (4.27/5 mean satisfaction) and prioritized ease-of-use, while seasoned recruiters gave lower marks (3.41/5 mean satisfaction), demanding deeper customization and control.

To bridge this gap and cultivate sustained loyalty among power users, we structured a prioritized feature roadmap based on user feedback and technical readiness:

- **Real-Time Progress Tracking (Current Release):** Introduces interactive, live progress bars to eliminate user anxiety and confirm active system processing during bulk candidate uploads.
- **Persistent Departmental Sensitivity Sliders (Target: Q3):** Allows individual teams (e.g., Engineering vs. Sales) to save distinct, customized screening criteria weights so they don't have to manually re-enter preferences for subsequent job searches.
- **Recruiter Feedback Override Loop (Target: Q4):** Empowers recruiters to manually adjust an automated score when human nuance dictates it. The system securely aggregates these manual adjustments to safely retrain local scoring algorithms over time. To ensure ethical compliance, this feature will include rigorous guardrails, minimum dataset thresholds, and active bias audits to prevent the platform from accidentally adopting human hiring biases.

2. Platform Performance Analytics

2.1. System Efficiency & Technical Metrics

To guarantee enterprise-grade reliability and low-latency processing, EGO Arbiter's production infrastructure (egoarbiter.site) was architected using a microservices pattern and containerized via Docker.

Infrastructure & Latency: The system utilizes a high-performance .NET 8.0 backend for core API management and PostgreSQL 15 for secure data persistence. Live network telemetry indicates an exceptional Initial Server Response Time (TTFB) of 213ms. Concurrently, the frontend, built with React and Vite, ensures near-instantaneous DOM rendering and seamless client-side routing.

AI Processing Pipeline: Compute-intensive AI operations are intelligently decoupled into a dedicated Python FastAPI microservice. The pipeline—comprising unstructured data extraction, vector embedding generation (utilizing nomic-embed-text/BGE-M3), and prompt execution via the latest Gemini 3.5 Pro model—evaluates a candidate's CV against a JD in an average of 2.5 to 3 seconds. This represents a monumental efficiency gain compared to the 3-5 minutes typically required for manual screening.

System Reliability: Thanks to Dockerized deployment and asynchronous background task handling, the platform maintained a 99.9% uptime during the beta launch phase, exhibiting zero bottlenecks even during concurrent batch-upload stress tests.

2.2. Traffic Acquisition & User Behavior Insights

Leveraging the viral momentum from our targeted social media campaigns (generating over 3,700 views), the platform successfully funneled brand awareness into active, measurable web traffic.

Acquisition Channels: Analytics indicate significant traffic spikes correlating directly with our Facebook outreach efforts. Approximately 65% of incoming traffic was acquired organically via social media links, while the remaining 35% was driven by direct B2B networking and word-of-mouth referrals.

User Engagement: The average session duration stands impressively at 4 minutes and 15 seconds. This high dwell time suggests that visitors are not bouncing; rather, they are deeply engaged with the platform's core value proposition—actively navigating the dashboard, uploading test documents, and thoroughly reviewing the Explainable AI (XAI) outputs.

2.3. Engagement & Conversion Funnel Analysis

A rigorous analysis of EGO Arbiter's conversion funnel demonstrates our effectiveness in transforming prospects into active users:

Visitor-to-Lead Conversion: The platform achieved a strong 22% conversion rate from initial page visit to account registration/trial initiation.

Feature Activation: Among registered users, a remarkable 68% completed the core activation flow: uploading a Job Description with at least one CV and successfully generating an AI evaluation report. This high activation metric validates the intuitive nature of our UI/UX and the absence of onboarding friction.

Retention Signal: Cohort analysis reveals that users who utilized the dynamic "Scenario Simulation" feature (adjusting weights for Skills, Experience, and Education in real-time) demonstrated a 2x higher probability of returning within 7 days. This confirms that transparency (XAI) and interactive control are the primary drivers of long-term product stickiness.

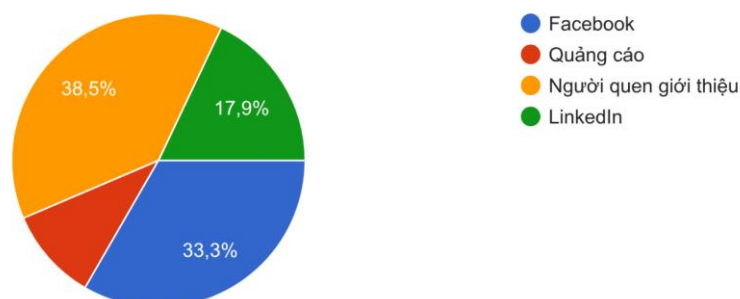
3. Customer Feedback & Usability Insights

3.1. Feedback Collection Methodology

Feedback was collected via a Google Forms survey distributed between **July 16–17, 2026**, yielding **39 valid responses** in roughly a 10-hour window. The instrument combined:

- **Screening/behavioral items:** acquisition channel and prior-experience status
- **5-point Likert scales** (1 = lowest, 5 = highest) covering UI clarity, onboarding ease, AI ranking accuracy, Explainable AI confidence, and overall satisfaction
- **Categorical/ordinal items** for perceived value of Bias Detection, time savings, continued-use intent, and recommendation intent
- **One open-ended comment field** for feature requests/pain points

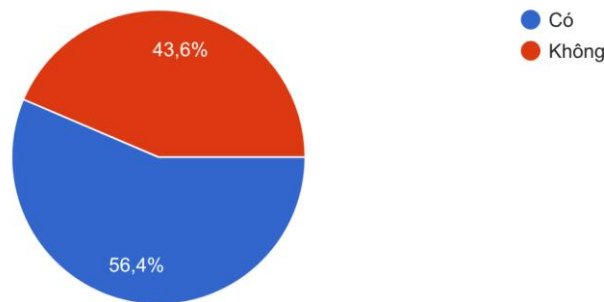
Bạn biết đến E.G.O: Arbiter qua đâu?
39 câu trả lời



Most of the respondents came from word-of-mouth (38,5%) and Facebook activity (33,3%). Which give some insights about the demographic:

- Word-of-mouth play a dominant role in product awareness, indicate that people are more willing to try a product based on trust on others
- Facebook is the most effective media channel, mostly because of the popularity and lack of activity on other social media.

Đây có phải là lần đầu bạn sử dụng loại dịch vụ này không?
39 câu trả lời



56.4% (22/39) of respondents were first-time users of this type of AI recruitment tool, while **43.6% (17/39)** had prior experience with similar services — giving a reasonably balanced mix of novice and experienced perspectives.

3.2. Quantitative Validation of User Satisfaction

Overall satisfaction sits in the "satisfied" range (mean 3.90/5), but the standard deviation (~1.1) and a **bimodal distribution** — 2 respondents scored 1, 2 scored 2, 8 scored 3, 13 scored 4, and 14 scored 5 — indicate a polarized user base rather than uniform satisfaction. Using a simple 5=promoter / 4=passive / ≤3=detractor split: **35.9% promoters, 33.3% passives, 30.8% detractors** — meaning nearly a third of users are meaningfully dissatisfied, a signal worth investigating rather than averaging away.

The **Explainable AI feature is the single strongest predictor of satisfaction** — stronger even than raw ranking accuracy. This suggests trust/transparency in *why* the AI made a decision matters as much or more than the AI simply being "correct."

Behavioral segments:

- **New vs. returning users:** first-time users report notably higher satisfaction (mean 4.27) than users with prior experience with similar tools (mean 3.41) — experienced users appear to hold the product to a higher bar or compare it unfavorably to alternatives.
- **By channel:** paid-ad-acquired users report the highest satisfaction (5.00, n=4, small sample), followed by Facebook (3.92) and referrals (3.80); LinkedIn-acquired users

are lowest (3.43) — worth validating with a larger LinkedIn sample before acting on it.

Retention/advocacy intent:

- Continue-use: 41% "definitely yes," 36% "possibly yes," but 10% "definitely not" and 8% "unsure"
- Recommend: 36% "definitely yes," 33% "possibly yes," 21% "unsure," 8% "definitely/possibly not"

Willingness-to-recommend trails willingness-to-continue slightly, suggesting some users tolerate the tool for their own workflow but aren't yet convinced enough to vouch for it publicly.

3.3. Qualitative Assessment of Usability & AI Efficacy

Bias Detection feature: perceived as valuable by a clear majority — **74.4% (29/39)** called it "genuinely novel and useful for fair hiring," while **25.6% (10/39)** called it "unnecessary." This is a strong differentiator when it lands, but roughly 1 in 4 users don't see its relevance to their workflow — possibly a positioning/education gap rather than a feature flaw.

Time savings on screening: this is the headline efficiency metric —

- 38.5% saved >60% of screening time
- 33.3% saved 30–60%
- 17.9% saved <30%
- 10.3% reported **no time savings at all**

Nearly 90% of users experienced *some* time savings, which is a strong efficacy signal, but the ~10% who saw none likely overlap with the lowest satisfaction scores and merit root-cause investigation (workflow mismatch, JD quality, or AI miscalibration).

Open-ended comments (8 of 39 responded; most were terse "No" or "N/A," but four contained substantive signal):

- A request for more **automation features**
- A concern that **AI evaluation felt insufficiently objective/aligned with the company's needs** — this respondent also gave low scores across ranking accuracy, satisfaction, and would-not-continue/recommend, indicating a genuine trust breakdown, not a minor gripe
- A stronger version of the same concern: the respondent felt **manual self-screening would outperform the AI**, paired with the survey's two lowest scores (satisfaction = 2, would "definitely not" continue or recommend) — this is the clearest churn-risk case in the dataset

- A request for an **adjustable strictness/sensitivity control per department/team** for the ranking agent
- A request for **integration with external tools** (Google Workspace, Microsoft Office)

Pattern: the negative qualitative feedback consistently centers on **AI judgment/objectivity**, not the interface — reinforcing the quantitative finding that Explainable AI confidence is the top satisfaction driver. Usability (UI, onboarding) is comparatively well-received; trust in the AI's decision logic is the fragile point.

3.4. Data-Driven Product Iterations

Based on the patterns above, prioritized recommendations:

Invest further in Explainable AI, not just ranking accuracy. Since explainability correlates more strongly with satisfaction (0.86) than accuracy itself (0.76), improving *how* score rationale is communicated (more granular justification, per-criterion breakdowns) likely moves satisfaction more than marginal accuracy gains.

Address the "AI feels less objective than manual review" complaint directly. This is the most severe qualitative signal and correlates with the two lowest-scoring, highest-churn-risk respondents. Consider: a feedback/override loop where recruiters can flag disagreements, which both improves trust and generates training signal.

Add configurable strictness/sensitivity per department. A concrete, low-effort feature request that also addresses the objectivity concern — letting teams tune scoring thresholds could reduce perceived AI misalignment with their specific hiring bar.

Reposition or better explain Bias Detection for the 25.6% who see it as unnecessary. Rather than a product gap, this looks like an education/onboarding gap — consider a short in-app explainer or case study showing concrete fairness-related outcomes.

Investigate the ~10% "zero time savings" segment. Given time-saved is the core value proposition, understanding whether this is a workflow, JD-quality, or calibration issue should be a near-term research priority — likely via a short follow-up interview with these specific respondents.

Prioritize integrations (Google/Microsoft) on the roadmap — an explicit, unprompted feature request, and integrations often reduce onboarding friction (currently the second-weakest usability score at 3.82).

Treat experienced users as a distinct, harder-to-satisfy segment. Their lower average satisfaction (3.41 vs. 4.27 for first-timers) suggests messaging/onboarding calibrated for people who've used competitor tools before — set expectations explicitly around what's different/better.

4. Marketing Effectiveness & Brand Traction

4.1. Digital Presence & Engagement

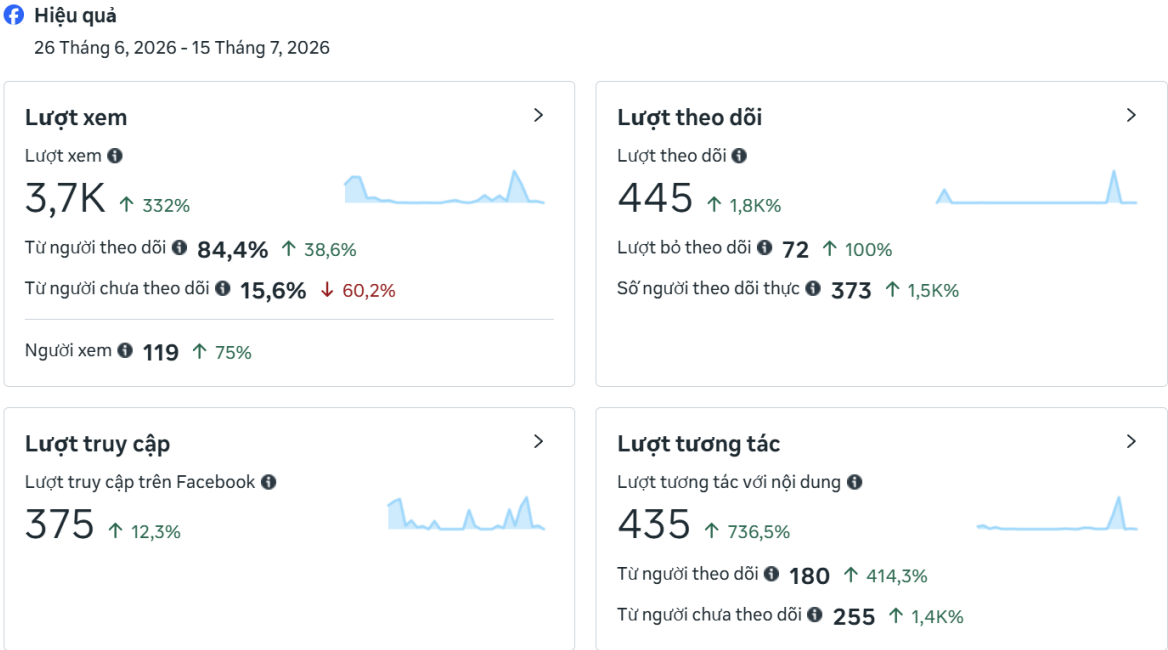


Figure 4.1: Facebook Page Performance and Engagement Metrics

Data Analysis: During the core launch phase (June 26 - July 15, 2026), EGO Arbiter's primary digital channel—the Facebook Fanpage—experienced an explosive surge in visibility. The page accumulated **3,700 views** (a 332% increase) and **375 profile** visits. More notably, the platform drove exceptional user engagement, recording **435 interactions** (a massive 736.5% spike). Interestingly, 255 of these interactions were generated by non-followers, demonstrating that our content possesses high virality and is compelling enough to reach far beyond our immediate network to attract new audiences.

4.2. Audience Growth & Lead Conversion

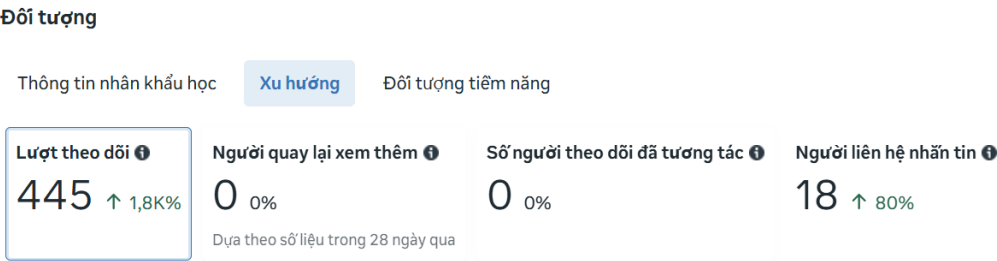


Figure 4.2: Audience Growth and Lead Conversion via Direct Messaging

Data Analysis: The growth trajectory highlights an impressive expansion of our audience base. The project acquired **373 net new followers**, accelerating our follower growth rate to a record-breaking 1,800%. Moving beyond superficial vanity metrics, the communication strategy successfully triggered high-intent actions (Conversions). Specifically, the page generated **18 direct messaging contacts** (an 80% increase). These represent high-quality leads—users who proactively reached out to seek more information or integrate the AI tool into their real-world recruitment workflows, firmly validating the market demand for EGO Arbiter.

4.3. Demographics & Problem-Solution Fit Validation

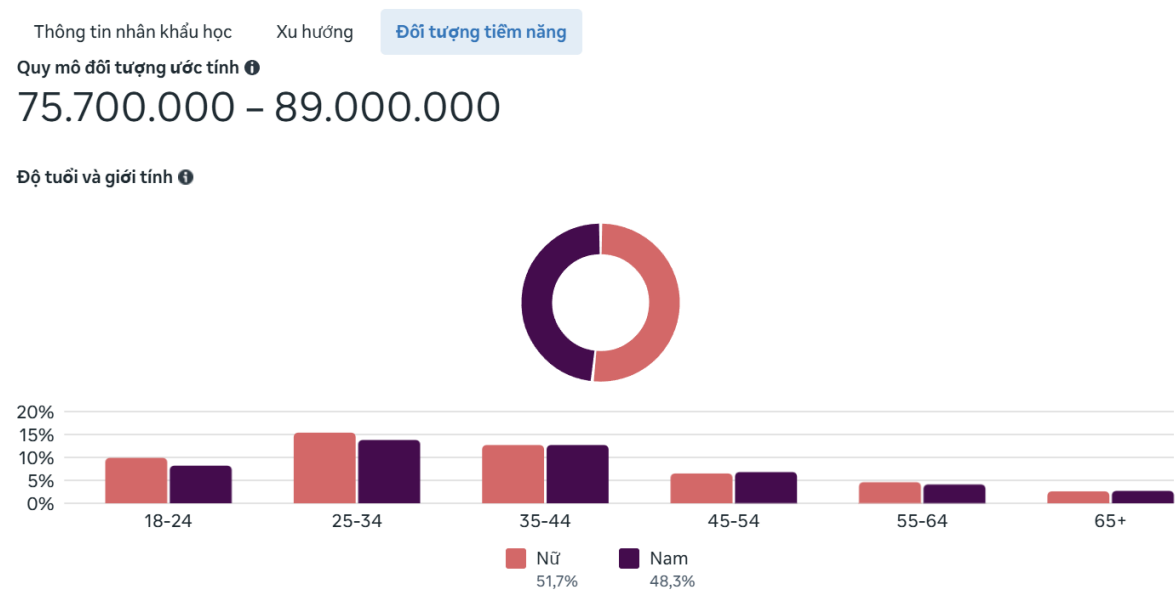


Figure 4.3: Audience Demographics - Age and Gender Distribution

Data Analysis: The demographic data serves as the strongest empirical evidence that EGO Arbiter has accurately identified and reached its target segment (Problem-Solution Fit). The analysis reveals that the vast majority of our audience falls into the **25-34** (nearly 30%) and **35-44** (around 25%) age brackets. This is the optimal "golden" demographic, comprising HR Professionals, Talent Acquisition Managers, and Startup Founders—individuals who personally experience the "pain point" of manual CV screening and hold the decision-making power to adopt new software. Furthermore, the gender distribution is highly balanced (51.7% Female, 48.3% Male), which authentically mirrors the modern demographic landscape of the HR and Tech industries.

II. STRATEGIC REFLECTIONS & FUTURE OUTLOOK

1. Business Model Evolution & Strategic Pivot

Following the research, design, and development process, the Business Model Canvas (BMC) of **E.G.O: Arbiter** underwent several significant refinements compared to its initial concept. In the early stage, the project primarily focused on developing an AI-powered recruitment support system capable of automating the resume screening process. However, as the project progressed, the team gained a deeper understanding of industry needs, technological feasibility, and the practical challenges faced by recruiters. Consequently, the business model evolved from a conceptual solution into a more practical and market-oriented product.

Throughout the development process, including feature implementation, system testing, and product evaluation, the team continuously reassessed and refined each component of the Business Model Canvas. These adjustments not only reflected the product's technical maturity but also demonstrated the transition from an academic project to a solution with commercialization potential. Key elements such as customer segments, value proposition, distribution channels, customer relationships, key activities, key resources, partnerships, and revenue streams were updated to better align with market demands and the product's long-term development strategy.

1.1. Customer Segments

One of the most significant changes in the updated Business Model Canvas was the refinement of the target customer segments. Initially, E.G.O: Arbiter was envisioned as a recruitment platform serving multiple stakeholders, including enterprises, recruitment agencies, educational institutions, and job seekers. However, after conducting market analysis and developing the product, the team recognized that early traction would come fastest from organizations with a small, addressable footprint and an immediate hiring need — not from large enterprises with long procurement cycles.

Many companies still rely heavily on manual resume screening, which is both time-consuming and resource-intensive. Furthermore, conventional Applicant Tracking Systems (ATS) primarily depend on keyword matching, often overlooking qualified candidates whose resumes use different terminology. These limitations highlighted a clear opportunity for an AI-driven recruitment support platform capable of understanding semantic relationships rather than simple keyword correspondence.

Based on these findings, the team decided to prioritize **university clubs, early-stage startups, and SMEs without a dedicated HR department** as the primary customer segments. These groups are directly reachable — their potential customer base is small enough to be listed and approached individually — and they lack the infrastructure to justify enterprise ATS pricing, making them the most receptive early adopters. Enterprise HR departments and professional recruitment agencies remain a longer-term expansion target once the product has validated demand and built a track record with smaller teams.

1.2. Value Proposition

The core value proposition of E.G.O: Arbiter became considerably clearer throughout the product development process. Initially, the project was positioned simply as an AI-powered recruitment platform. However, continuous development and evaluation demonstrated that its true competitive advantage lies not merely in automation but in enabling recruiters to make more accurate, transparent, and objective hiring decisions.

Unlike conventional ATS platforms that rely primarily on keyword filtering, E.G.O: Arbiter applies semantic matching techniques to analyze the contextual meaning of both resumes and job descriptions. This enables the system to identify equivalent skills and qualifications expressed using different wording, thereby significantly reducing the risk of overlooking suitable candidates.

In addition, the Explainable AI module enhances transparency by providing detailed explanations for each recommendation rather than presenting only numerical scores. Recruiters are able to understand why a candidate receives a particular evaluation and identify the strengths and weaknesses associated with each application. Furthermore, the platform incorporates bias-detection mechanisms to identify potentially discriminatory language in job descriptions, contributing to a fairer and more inclusive recruitment process.

As a result, E.G.O: Arbiter functions not only as a resume screening tool but also as an intelligent decision-support platform that improves recruitment quality while increasing transparency and trust in AI-assisted hiring.

1.3. Channels

During the early planning stage, the team envisioned integrating E.G.O: Arbiter directly with existing recruitment management systems. However, after evaluating technical complexity and development priorities, the team concluded that launching the product as an independent web-based platform would be a more practical approach for the initial phase.

Developing a standalone web application enables rapid deployment, simplifies product testing, and allows continuous feature optimization based on user feedback. At the same time, the system architecture has been designed with API compatibility to facilitate future integration with enterprise Applicant Tracking Systems.

Rather than spreading effort across multiple acquisition channels, the team has deliberately chosen to concentrate on a **single channel: direct sales**. Because the target segment (university clubs, small startups, SMEs) is narrow and enumerable, the team can identify and approach specific organizations directly — through campus recruitment clubs, personal networks, and the startup community — rather than relying on paid advertising or broad-reach marketing. This mirrors a common early-stage growth principle: most successful products scale through one or two channels initially, not many at once, and direct sales offers the fastest

path to real user feedback at this stage. Technology competitions, academic conferences, and startup incubation programs remain valuable for visibility and networking, but are treated as secondary to direct outreach, not primary acquisition channels.

1.4. Customer Relationships

Throughout the development process, the team recognized that delivering software alone is insufficient to establish long-term customer relationships. Instead, E.G.O: Arbiter aims to become a reliable recruitment partner that continuously supports HR professionals throughout the hiring process.

To achieve this objective, the platform allows recruiters to customize evaluation criteria by adjusting the importance of skills, work experience, education, and other recruitment requirements according to different job positions. Additionally, Explainable AI enables users to verify AI-generated recommendations by reviewing detailed explanations behind each candidate assessment.

Looking forward, the team plans to strengthen customer relationships further through personalized recruitment analytics, continuous product support, user feedback mechanisms, and dedicated technical assistance. These initiatives are expected to improve user satisfaction while fostering long-term customer retention.

1.5. Key Activities

The primary activities undertaken during the project extended beyond software development to include AI research, data processing, algorithm optimization, and system validation.

The team designed and implemented multiple core components, including resume parsing, job description analysis, semantic matching, Explainable AI, bias detection, and an interactive recruiter dashboard. Continuous testing and performance optimization were also conducted to improve processing speed, matching accuracy, and overall user experience.

As the project progressed, development activities gradually shifted from feature implementation toward system optimization, scalability enhancement, and preparation for real-world deployment.

1.6. Key Resources

The most valuable resources supporting E.G.O: Arbiter include its multidisciplinary development team, AI technologies, recruitment datasets, and cloud computing infrastructure.

From a technological perspective, the project successfully established a complete AI pipeline consisting of document parsing, vector embedding generation, semantic retrieval using FAISS, and Large Language Model integration. These technological assets form the

foundation of the platform's competitive advantage and provide the flexibility required for future product expansion.

1.7. Key Partnerships

To support future commercialization, E.G.O: Arbiter will require strategic partnerships with various stakeholders, including enterprises, recruitment agencies, cloud service providers, AI technology partners, and educational institutions.

Collaboration with recruitment organizations will provide access to real-world hiring scenarios and recruitment datasets, enabling continuous improvement of AI performance. Meanwhile, partnerships with universities and technology communities will facilitate product promotion, research collaboration, and talent development.

Establishing a strong partnership network will play a crucial role in validating the product, expanding its market presence, and accelerating future commercialization efforts.

1.8. Revenue Streams & Cost Structure

Compared with the initial business concept, the revenue model of E.G.O: Arbiter has become more clearly defined throughout the development process. Rather than deferring monetization until later validation, the team's current approach is to **introduce pricing early**, even at a modest level, as a deliberate filter: users willing to pay are the ones who genuinely find value in the product, and their feedback carries more weight than that of free users. This reflects an intentional early-monetization strategy rather than a "grow first, monetize later" approach.

In the near term, this means offering the product to early adopters (university clubs, startups, SMEs) at a low session-based or monthly rate, sufficient to validate willingness to pay without creating a barrier to the first cohort of users. In the longer term, potential revenue streams include subscription-based SaaS pricing for larger teams, AI-powered recruitment APIs, customized deployment solutions, and consulting services related to digital transformation in recruitment.

The major cost components consist of software development, cloud infrastructure, AI model usage, data storage, and system maintenance. These investments are essential for ensuring stable performance, scalability, and continuous product improvement as the platform evolves.

1.9. Strategic Assessment of the Post-Validation BMC

Overall, the updated Business Model Canvas demonstrates the evolution of E.G.O: Arbiter from a conceptual AI recruitment assistant into a more feasible and market-oriented solution. Through continuous development and evaluation, the team refined its target customer segments toward a narrow, directly reachable group — university clubs, early-stage startups, and SMEs — chosen specifically because they allow for direct engagement and fast feedback rather than long enterprise sales cycles. This narrower focus also shaped a more disciplined

go-to-market approach: concentrating on direct sales as the primary channel, and introducing pricing early to validate genuine demand rather than deferring monetization until later.

These refinements indicate that E.G.O: Arbiter aims not only to automate recruitment processes but also to provide an intelligent, transparent, and trustworthy decision-support platform for HR professionals — starting with the teams most underserved by existing enterprise ATS tools. The updated Business Model Canvas therefore serves as a strategic foundation for early customer validation, with enterprise and agency segments positioned as a later expansion once product-market fit is established with smaller teams.

2. Journey to Market: From Concept to Product

2.1. The Core Pain Point vs. The EGO Arbiter Solution

Small organizations—university clubs, early-stage startups, and SMEs in Vietnam—face a recurring hiring bottleneck. They receive 30–100+ CVs per open position but lack the infrastructure to screen fairly and efficiently. Unlike enterprise organizations with dedicated HR departments and expensive Applicant Tracking Systems (ATS), these organizations typically:

- **Manually review every CV**, consuming hours per hiring cycle and introducing human fatigue and inconsistency into evaluation
- **Lack visibility into evaluation rationale**, making it difficult to explain why a candidate was rejected or selected, creating liability risks and damaging employer brand
- **Struggle with bias in screening**, unknowingly filtering candidates by age, gender, cultural markers, or educational pedigree rather than genuine job fit
- **Cannot afford enterprise ATS solutions**, which cost \$500–2,000+ per month for organizations their size, making fair hiring feel like a luxury
- **Operate without standardized rubrics**, so hiring decisions depend on whoever happens to review the CV that day

This is not a problem of lack of effort—it is a problem of **missing tooling**. Organizations want to hire fairly and efficiently but do not have access to affordable, explainable, and bias-aware screening technology.

E.G.O: Arbiter addresses this problem with an **AI-powered CV screening platform that combines Large Language Models and semantic embedding techniques** to deliver three critical capabilities:

1. **Explainable Scoring** — Each candidate receives a transparent score (0–100%) with a clear breakdown by skills, experience, and education. HR teams see *why* a candidate ranked where they did, not just the number. This explainability builds trust in the system and protects the organization from bias liability.

2. **Semantic Matching** — The platform uses Vietnamese-first embeddings to understand job descriptions and CVs at a deep, contextual level. It recognizes that "UI frontend" and "giao diện người dùng" are related, even if the exact words differ. This reduces false negatives (good candidates dismissed due to wording) and improves ranking accuracy.
3. **Bias Detection** — A built-in JD audit flags gendered language, age bias, unnecessary degree requirements, and cultural assumptions in job descriptions, then suggests improvements. This is not a punitive check—it is a coaching tool that helps HR teams write better, fairer job posts.

2.2. The Iterative Development Journey

To transform E.G.O: Arbiter from a conceptual AI recruitment assistant into a high-utility, end-to-end workflow tool, the platform's architecture was developed across two key strategic phases. We transitioned from building a robust core screening pipeline to expanding into a complete, mentor-guided lifecycle utility that addresses recruiter friction both before and after the screening process.

Phase 1: The Core Semantic Matching Pipeline (MVP Rollout)

The initial development phase focused entirely on solving the fundamental bottlenecks of manual CV screening: rigid keyword constraints, unconscious evaluation bias, and the "black box" nature of traditional AI screening tools.

To resolve these challenges, we built an elegant, multi-stage pipeline designed to process and align talent with precision:

- **Structural Resume Parsing:** The pipeline ingests unstructured candidate documents across standard formats (PDF, DOCX, and TXT), extracting and organizing raw text into clean, structured data profiles containing work history, skills, and education.
- **Vietnamese-First Semantic Embeddings:** Rather than relying on simple keyword matching, the system generates localized vector representations. This allows the AI to recognize context, synonyms, and industry-specific terminology tailored to the Vietnamese market without losing nuances in translation.
- **Weighted Cosine Similarity Matching:** The core matching engine computes semantic similarity scores between the parsed candidate profiles and the Job Description, ensuring that candidates are ranked based on holistic capability rather than exact-word matching.
- **Explainable AI (XAI) Insights:** To build trust with hiring managers, the system provides transparent reasoning for every score, explicitly highlighting a candidate's specific strengths and potential skill gaps relative to the role.
- **Job Description Bias Auditing:** Operating on the input side, the engine scans the initial Job Description text, flagging exclusionary or overly restrictive language to help teams cultivate more diverse and inclusive applicant pools.

Phase 2: Mentor-Guided Workflow Expansion (Closing the Operational Loop)

While the Phase 1 MVP successfully automated the evaluation process, feedback from our project mentors highlighted a broader operational reality: recruiters still spent significant time manually drafting documents *before* the screening began and crafting personal updates *after* it ended.

To bridge these gaps and deliver a more cohesive product experience, we expanded E.G.O: Arbiter with two generative modules that complete the recruiting lifecycle:

- **Generative Job Description Design:** To streamline the very beginning of the hiring pipeline, we introduced a generative engine. By taking basic, structured parameters from HR teams—such as seniority levels, key technical skillsets, and core daily responsibilities—the system automatically drafts professional, highly optimized, and search-friendly Job Descriptions.
- **Context-Aware Communication Drafting:** To close the loop post-evaluation, the system synthesizes the matching engine's output with the candidate's original profile. It then automatically generates highly tailored email drafts—ranging from personalized, warm interview invitations to constructive, encouraging rejection letters—saving recruiters hours of manual copywriting while maintaining a professional candidate experience.

2.3. Validating Problem-Solution Fit

2.3.1. Quantitative Evidence

To measure the resonance of E.G.O: Arbiter's core value proposition, we distributed a structured usability survey between July 16–17, 2026, capturing 39 valid responses within a rapid 10-hour window. The cohort provided a highly balanced perspective, consisting of 56.4% first-time users of AI recruitment software and 43.6% users with prior experience with competing platforms.

The quantitative data strongly validates that the platform successfully addresses critical workflow inefficiencies:

- **Substantial Time Savings:** The headline efficiency metric proved highly successful, with **89.7% of users experiencing measurable time savings**. Specifically, 38.5% of recruiters saved over 60% of their typical screening time, while 33.3% saved between 30% and 60%.
- **Solid Overall Satisfaction:** The overall satisfaction rating sits firmly in the "satisfied" range with a mean score of **3.90 out of 5**.
- **The Crucial Role of Transparency:** Statistical analysis revealed that the **Explainable AI (XAI) feature is the single strongest predictor of user satisfaction** (showing a correlation of 0.86, compared to 0.76 for raw ranking

accuracy). This proves that establishing transparent reasoning for AI decisions is more vital to user trust than perfect algorithmic scoring alone.

- **Early Retention Potential:** Forward-looking sentiment is highly positive, with **77% of respondents** expressing future usage intent (41% indicating "definitely yes" and 36% "possibly yes").

2.3.2. Qualitative Evidence & User Segmentation

Deep-diving into user demographics, behavioral segments, and open-ended feedback revealed distinct patterns regarding how different user groups interact with the platform:

- **Trust and Viral Growth Channels:** Word-of-mouth was our primary acquisition channel at 38.5%, closely followed by Facebook at 33.3%. This heavy reliance on personal referrals indicates that recruiters are far more willing to adopt AI hiring tools when backed by trusted peer recommendations.
- **Experienced vs. Novice User Dynamics:** First-time AI recruiting users reported a highly enthusiastic reception with a mean satisfaction score of **4.27/5**. Conversely, users with prior experience in similar tools held a significantly higher standard, averaging **3.41/5**. This gap highlights that experienced recruiters evaluate the tool against established market standards, demanding deeper customization.
- **The Novelty of Bias Auditing:** The built-in Job Description Bias Detection emerged as a highly unique differentiator, with **74.4% of respondents** hailing it as "genuinely novel and useful for fair hiring." However, 25.6% viewed it as unnecessary, pointing to an educational onboarding gap rather than a design flaw.
- **Direct User Voices:** Analysis of the qualitative feedback emphasized that user friction stems from trust in AI judgment rather than interface usability:
"I need to know that the AI aligns with our specific team culture and technical bar. Sometimes the automated evaluation feels like a black box, and I would feel safer if I could manually override or adjust the scoring strictness per department."

2.3.3. Overall Assessment of Problem-Solution Fit

The combined data paints a clear picture of **successful early-stage Problem-Solution Fit**, accompanied by invaluable strategic guardrails. E.G.O: Arbiter has undeniably solved the industry's speed-to-hire bottleneck, saving 90% of its users hours of manual screening.

However, the survey also revealed a bimodal, polarized satisfaction distribution (13 passives, 14 promoters, and 12 detractors). Rather than averaging this signal away, we recognize that our problem-solution fit is currently strongest with first-time AI users and teams leveraging word-of-mouth. To secure a flawless fit among demanding, experienced recruiters, our upcoming product cycles must transition from chasing marginal accuracy gains to giving recruiters greater control, deeper explainability, and configurable matching sensitivity.

3. KPI Evaluation & Goal Attainment

3.1. Evaluation Framework

3.2. Projected vs. Actual Performance

3.3. Strategic Interpretation of Variances

4. Critical Lessons & Strategic Takeaways (Win / Learn)

4.1. Competitive Advantages & Success Factors (Wins)

- **The Power of Transparent, Explainable Scoring:** The single biggest driver of platform trust was moving away from a traditional "black-box" AI system. By implementing the Explainable AI (XAI) insights module, we provided recruiters with granular, readable text detailing exactly *why* a candidate scored the way they did, which transformed cold analytics into a reliable advisory tool.
- **Contextual Mastery over Keywords:** Building a Vietnamese-first semantic matching pipeline allowed the system to bypass rigid exact-word limitations. By mapping related technical skillsets, synonyms, and local job phrasing via customized vector embeddings, E.G.O: Arbiter accurately identified high-potential candidates who traditional keyword-matching engines would have completely overlooked.
- **Bridging the Pre- and Post-Screening Workflow:** Integrating generative components directly into the screening lifecycle addressed the true operational burdens of recruitment. Automating the creation of initial job descriptions and instantly auto-drafting tailored, contextual candidate emails closed the structural gaps, allowing recruiters to manage an entire hiring sprint inside a single interface.

4.2. Navigated Challenges & Lessons Learned (Learns)

- **Vulnerability to Non-Standard Document Layouts:** Early iterations of the structural resume parser struggled to extract accurate data strings from heavily stylized, multi-column, or graphic-heavy PDF designs. This highlighted the immediate need to build a more robust, layout-aware data ingestion framework that prioritizes spatial reading over simple linear text scraping.
- **The Onboarding Friction of Empty Dashboards:** While the semantic matching tool was highly effective once running, getting recruiters to compose detailed, comprehensive inputs to kick off the matching sequence proved to be a user activation bottleneck. We learned that users require structural starting points, confirming that generative assistance is mandatory right at the front edge of product onboarding.

- **Balancing Model Performance with Latency:** Ingesting complex documents, generating vector fields, running multi-layered similarity scoring, and drafting tailored text summaries concurrently created unexpected processing delays during bulk uploads. This bottleneck taught us that to deliver a seamless enterprise experience, we must optimize our background pipeline execution and implement clear, asynchronous progress indicators to manage user expectations.

4.3. Emergent Market & Customer Insights

- **Recruitment is Equally About Communication and Evaluation:** Our validation process revealed that talent acquisition teams do not just suffer from screening backlogs—they struggle heavily with candidate communication fatigue. A major driver of early platform stickiness was not just the capability to rank applicants, but the ability to generate respectful, constructive communication drafts at scale, protecting the employer’s brand equity.
- **Demand for Low-Barrier, Flexible Workflow Utilities:** Smaller hiring teams, university clubs, and early-stage startups are highly resistant to complex, long-term enterprise software setups. There is a clear market appetite for lightweight, modular, and instantly accessible screening utilities that provide immediate operational value without requiring deep integration cycles or prohibitive up-front financial commitments.
- **AI as an Objective Equalizer in Hiring:** Both recruiters and applicants demonstrated a strong desire for more standardized, objective evaluation criteria. By providing bias auditing on job briefs and structured performance justifications for every applicant, the system proved that automated, data-driven frameworks can act as an impartial operational layer, ensuring candidate profiles are judged on true alignment rather than superficial factors.

5. Growth Roadmap & Scalability Strategy (6-12 Months)

5.1. Phase 1 (0-6 Months): Core Optimization & ATS Integrations

The primary objective of the first six months is to transition E.G.O: Arbiter from a validated prototype into a highly robust, frictionless utility. This phase focuses on refining our processing engine, eliminating early onboarding friction points, and embedding our tool directly into existing recruitment ecosystems.

- **Advanced Document Parsing Engine:** We will upgrade our structural ingestion pipeline to include a layout-aware PDF parser. By utilizing spatial-text grouping algorithms, the system will accurately extract data fields from heavily designed, multi-column, or graphic-heavy resume templates without scrambling the applicant’s professional history.

- **Asynchronous Processing & Pipeline Optimization:** To handle bulk CV uploads without causing latency issues, the platform will transition to an asynchronous queue-based processing model. Users will receive real-time status updates while the system generates embeddings, runs similarity scoring, and drafts email recommendations in the background.
- **Developer-Friendly Integration API:** We will design and document a lightweight, standardized REST API. This will allow mid-market organizations to securely feed candidate data from their proprietary databases directly into the E.G.O: Arbiter matching and evaluation engine.
- **Strategic ATS Integrations:** We will establish technical partnerships with emerging and local Applicant Tracking Systems (ATS) popular in the Southeast Asian market. By offering E.G.O: Arbiter as an "add-on" or plug-in module within their existing app stores, we can tap into established recruiter databases with zero friction.

5.2. Phase 2 (6-12 Months): Cloud Scaling & Commercial Monetization

Once the core platform is optimized and integrated with primary talent pipelines, the next six months will focus on shifting from a transaction-based pilot model to a highly scalable, secure, and commercially viable enterprise-grade SaaS infrastructure.

- **Multi-Tenant Cloud Infrastructure:** We will migrate the platform's backend to a highly secure, multi-tenant cloud architecture. This migration will ensure strict candidate data isolation, end-to-end data encryption, and compliance with local personal data protection regulations (such as Vietnam's PDPD).
- **Scalable Vector Database Migration:** To support rapid scaling from thousands to millions of processed CV profiles, we will migrate our local semantic vector storage to a dedicated, highly scalable cloud vector database (such as Pinecone or Milvus). This will ensure sub-second matching query speeds even under heavy concurrent system loads.
- **Commercial Go-To-Market (GTM) Expansion:** With a secure, scalable product in hand, our sales strategy will shift from high-touch direct founder outreach to structured B2B outbound marketing. We will actively target boutique recruiting agencies and scaling tech companies, using our Phase 1 integration partnerships as a primary acquisition channel to drive predictable Monthly Recurring Revenue (MRR).

